EE6094
CAD for VLSI Design

Checker

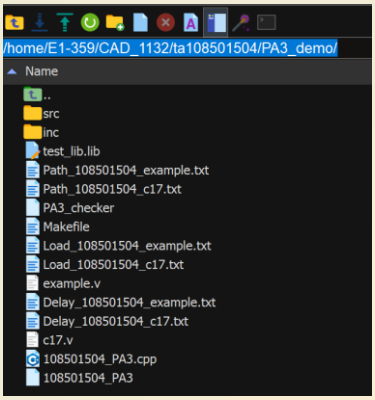
Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

Checker

- What is the checker?
  - Help you to check whether your solution is correct.
- Checker for PA3
  - Step1 ➔ Check the Output load
  - Step2 ➔ Check the Gate delay information
  - Step3 ➔ Check the longest and shortest path

Checker

- How to use the checker?
  - Step1: Make sure you put your checker, input case, liberty file and your output file Load, Delay, Path at the same folder.



Checker

- How to use the checker?
  - Step2: Key in the following commands to run the checker.

./PA3_checker *case test_lib.lib Load.txt Delay.txt Path.txt*

```
[ta108501504@ncuee ~/PA3_demo]$ ./PA3_checker c17.v test_lib.lib Load_108501504_c17.txt Delay_108501504_c17.txt Path_108501504_c17.txt
```

Checker

- If you get the **Permission denied** error, key in the following commands and try step2 again.

  chmod +x PA3_checker

```
[ta108501504@ncuee ~/PA3_demo]$ ./PA3_checker c17.v test_lib.li
./PA3_checker: Permission denied.
[ta108501504@ncuee ~/PA3_demo]$ chmod +x PA3_checker
```


Checker

- c17 result example.

```
Step 1 Correct!!!
Step 2 Correct!!!
Step 3-1 Correct!!!
Step 3-2 Correct!!!
Score: 8
```

Checker

- However, there are some errors which will cause your result wrong.
- We will give you an error message file for you to debug.

```
[ta108501504@ncuee ~/PA3_demo]$ ls
108501504_PA3      c17.v                    example.v  Load_108501504_c17.txt  PA3_checker            src
108501504_PA3.cpp  Delay_108501504_c17.txt  inc        Makefile                Path_108501504_c17.txt  test_lib.lib
[ta108501504@ncuee ~/PA3_demo]$ ./PA3_checker c17.v test_lib.lib Load_108501504_c17.txt Delay_108501504_c17.txt Path_108501504_c17.txt
Step 1 Correct!!!
Step 2 Correct!!!
Step 3-1 Correct!!!
Step 3-2 Correct!!!
Score: 8
[ta108501504@ncuee ~/PA3_demo]$ ls
108501504_PA3      c17.v                    error_message_c17.txt  inc                     Makefile     Path_108501504_c17.txt  test_lib.lib
108501504_PA3.cpp  Delay_108501504_c17.txt  example.v              Load_108501504_c17.txt  PA3_checker  src
[ta108501504@ncuee ~/PA3_demo]$
```


Checker

- Step1 Wrong!

```
[ta108501504@ncuee ~/PA3_demo]$
Step 1 Wrong!!!
Step 2 Correct!!!
Step 3-1 Correct!!!
Step 3-2 Correct!!!
Score: 5
[ta108501504@ncuee ~/PA3_demo]$
```

```
==================================================================
                              Step 1
==================================================================
Mismatch at line 2:
  Expected: U11 0.030000
  Actual:   U13 0.018896
Mismatch at line 3:
  Expected: U13 0.018896
  Actual:   U9 0.010811
Mismatch at line 4:
  Expected: U9 0.010811
  Actual:   U10 0.010501
Mismatch at line 5:
  Expected: U10 0.010501
  Actual:   U12 0.007985
Mismatch at line 6:
  Expected: U12 0.007985
  Actual:   U14 0.006836
Line count mismatch:
  Expected lines: 7
  Actual lines:   6
==================================================================
                              Step 2
==================================================================
==================================================================
                              Step 3
==================================================================
```

Checker

◆Step2 Wrong!

```
[ta108501504@ncuee ~/PA3_demo]$
Step 1 Correct!!!
Step 2 Wrong!!!
Step 3-1 Correct!!!
Step 3-2 Correct!!!
Score: 5
```

```
==========================================
                    Step 1
==========================================
==========================================
                    Step 2
==========================================
Mismatch at line 2:
  Expected: U8 1 0.064212 0.103237
  Actual:   U8 0 0.064212 0.103237
Mismatch at line 3:
  Expected: U12 1 0.044436 0.059305
  Actual:   U12 1 0.02 0.059305
Mismatch at line 6:
  Expected: U9 1 0.028109 0.053694
  Actual:   U9 1 0.028109 0.0122
Line count mismatch:
  Expected lines: 7
  Actual lines:   6
==========================================
                    Step 3
==========================================
```


Checker

◆Step3 Wrong!

```
[ta108501504@ncuee ~
t Delay_108501504_c1
Step 1 Correct!!!
Step 2 Correct!!!
Step 3-1 Wrong!!!
Step 3-2 Wrong!!!
Score: 6
```

```
==========================================
                    Step 3
==========================================
Longest delay mismatch!!!
  Golden: 0.158793
  Calculated: 0.258793
Longest path connection mismatch!!!
  n10 -> N22
Longest delay and Path delay mismatch!!!
  Your longest delay: 0.258793
  Your path delay: 0.158793
Shortest path connection mismatch!!!
  n10 -> N23
~
```